

5G FUNDAMENTALS AND ARCHITECTURES

EC23712

ASSIGNMENT 1

Team 6

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Implementation of Sustained 5-Minute TCP Transfer and Rate Stability/Jitter Analysis Using Open5GS

1. Objective

The objective of this experiment is to evaluate the long-duration performance of the Open5GS 5G Core Network by generating a single TCP uplink flow for 300 seconds (5 minutes) using iperf3. The throughput values obtained throughout the experiment are analyzed to determine the mean throughput, peak throughput, minimum throughput, and throughput stability (jitter). The results are visualized using Python and Matplotlib to study the rate stability over time.

2. Software and Tools Used

Software	Purpose
Ubuntu 22.04	Operating System
Open5GS	5G Core Network
MongoDB	Subscriber Database
Node.js 20	Open5GS WebUI Support
Open5GS WebUI	Subscriber Management
iperf3	TCP Traffic Generation
Python 3	Data Processing
Matplotlib	Throughput Visualization
NumPy	Statistical Analysis

3. System Setup

3.1 Network Architecture

The experiment was carried out on a single Ubuntu system running the complete Open5GS core network.

The setup consists of:

- Open5GS Control Plane Functions

- NRF
- SCP
- AMF
- SMF
- AUSF
- UDM
- PCF
- NSSF
- BSF
- UDR
- Open5GS User Plane Function (UPF)
- MongoDB Subscriber Database
- Open5GS WebUI
- TUN Interface (ogstun – 10.45.0.1/16)
- iperf3 TCP Server and Client
- Python with Matplotlib for Visualization

4. Procedure

The following steps were performed during the experiment:

1. Installed MongoDB and configured it with Open5GS.
2. Built Open5GS from source and verified the successful compilation of all 5G core network functions.
3. Created and configured the ogstun TUN interface for user-plane communication.
4. Started all Open5GS core network functions by launching the 5GC application.
5. Started the iperf3 server to receive TCP traffic.
6. Generated a single-flow TCP uplink traffic for 300 seconds (5 minutes) using iperf3.
7. Saved the throughput measurements in JSON format as `sustained_result.json`.

8. Developed a Python script using Matplotlib and NumPy to process the recorded throughput data.
9. Plotted the throughput versus time graph, including the mean throughput, standard deviation (jitter), and jitter band for performance analysis.
10. Analyzed the sustained transfer results by evaluating the mean throughput, peak throughput, minimum throughput, and throughput stability (jitter) over the entire 5-minute duration.

5. Python Program

```
import json

import matplotlib.pyplot as plt

import numpy as np

with open("sustained_result.json") as f:
    data = json.load(f)

times = [iv["sum"]["start"] for iv in data["intervals"]]
mbps = [iv["sum"]["bits_per_second"] / 1e6 for iv in data["intervals"]]

mean = np.mean(mbps)
jitter = np.std(mbps)

plt.figure(figsize=(14, 5))
plt.plot(times, mbps, color="blue", label="TCP Throughput")
plt.axhline(mean, color="orange", linestyle="--", label=f"Mean: {mean:.0f} Mbps")
plt.axhline(mean + jitter, color="green", linestyle=":", label=f"Mean+Std: {mean+jitter:.0f} Mbps")
plt.axhline(mean - jitter, color="red", linestyle=":", label=f"Mean-Std: {mean-jitter:.0f} Mbps")
plt.fill_between(times, mean - jitter, mean + jitter, alpha=0.1, color="gray",
label="Jitter Band")
plt.xlabel("Time (s)")
```

```
plt.ylabel("Throughput (Mbps)")
plt.title("Team 6 - Sustained 5-Min TCP Transfer: Rate Stability & Jitter")
plt.legend()
plt.grid(True)
plt.savefig("team6_sustained.png")
plt.show()
```

```
print(f"Mean Throughput : {mean:.2f} Mbps")
print(f"Jitter (Std Dev): {jitter:.2f} Mbps")
print(f"Max          : {max(mbps):.2f} Mbps")
print(f"Min          : {min(mbps):.2f} Mbps")
```

6. Results

6.1 Network Performance Metrics

Performance Parameter	Value
Test Duration	300 seconds (5 min)
Protocol	TCP
Flow Type	Single Flow
Mean Throughput	45,582.65 Mbps
Peak Throughput	64,804.00 Mbps
Minimum Throughput	19,966.09 Mbps
Jitter (Standard Deviation)	7,088.02 Mbps

Figure 1: Network Performance Metrics Table

6.1 Throughput Stability Graph

The throughput variation over the entire 300-second TCP transfer is shown below.

The graph illustrates:

- Instantaneous TCP throughput
- Mean throughput
- Mean + Standard Deviation
- Mean – Standard Deviation
- Jitter band



Figure 1: Sustained 5-Minute TCP Transfer – Rate Stability and Jitter

6.3 Observations

The sustained TCP transfer maintained an average throughput of approximately 45.58 Gbps throughout the five-minute experiment.

The throughput initially reached a peak value of 64.80 Gbps, indicating efficient utilization of available bandwidth during the beginning of the transfer.

Throughout the experiment, throughput fluctuated due to TCP congestion control mechanisms, buffer management, and dynamic transmission window adjustments.

Most throughput values remained within the shaded jitter band, showing relatively stable network behaviour despite periodic variations.

The minimum throughput observed was 19.97 Gbps, occurring during temporary congestion periods. However, the throughput quickly recovered, demonstrating the robustness of the TCP protocol in adapting to changing network conditions.

The calculated standard deviation of 7.09 Gbps represents the throughput jitter. A moderate jitter value indicates that the network maintained good stability during long-duration data transfer while still experiencing expected TCP throughput oscillations.

Overall, the Open5GS 5G core sustained consistently high throughput for the complete five-minute duration without any interruption.

7. Analysis

The throughput graph demonstrates the dynamic behaviour of TCP over an extended transfer period.

The orange dashed line represents the average throughput of 45,582.65 Mbps, while the green and red dotted lines represent one standard deviation above and below the mean.

The shaded gray region corresponds to the throughput stability range (jitter band). Most throughput measurements fall within this region, indicating stable performance.

Several temporary spikes above 60 Gbps and occasional drops below 30 Gbps were observed. These variations are expected because TCP continuously adjusts its congestion window according to acknowledgements, network buffering, and available bandwidth.

Unlike short-duration tests, the five-minute sustained transfer provides a more realistic representation of long-term network performance, making it suitable for evaluating stability and reliability.

8. Conclusion

The experiment successfully evaluated the long-term performance of the Open5GS 5G Core Network using a sustained five-minute TCP transfer.

The measured average throughput was 45.58 Gbps, with a maximum throughput of 64.80 Gbps and a minimum throughput of 19.97 Gbps.

The throughput remained consistently high throughout the experiment despite normal TCP fluctuations. The calculated jitter of 7.09 Gbps indicates moderate

throughput variation and demonstrates that the Open5GS platform can sustain stable high-speed TCP communication over extended durations.

Overall, the experiment confirms that Open5GS provides a reliable environment for evaluating long-duration TCP traffic performance and analyzing throughput stability in software-based 5G networks.

9. References

1. Open5GS Documentation

<https://open5gs.org/open5gs/docs/guide/02-building-open5gs-from-sources/>

2. iperf3 Documentation

<https://iperf.fr/>

3. Open5GS GitHub Repository

<https://github.com/open5gs/open5gs>